

Article

Development of a Smart City Platform Based on Digital Twin Technology for Monitoring and Supporting Decision-Making

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Abstract: Information and communication technology's role in developing smart city platforms has allowed cities to grow smarter in recent years. In order to develop a smart city platform, digital twin technology can be implemented to monitor and simulate the city's conditions. Furthermore, it can function as a precise decision-support system. Digital twins can be combined with augmented reality technology to develop a smart city platform. The combination of these two technologies aims to visualize data for monitoring and simulating conditions in a city. The primary concern about the necessity of a digital twin platform in smart cities pertains to creating a robust digital twin-enabled smart city platform that can efficiently monitor urban conditions and provide significant insights for decision-making. Hence, this research aims to develop a smart city platform with digital twins as its foundation. This platform would enable real-time data visualization inside an environment that facilitates clear and effective information communication to users. The smart city platform development method is divided into four layers, namely developing (1) the basic layer that contains basic information about the city; (2) the 3D layer that contains the city's 3D assets; (3) the digital twin layer for real-time data integration; (4) the augmented layer for augmenting the digital twin data. This research also proposes an architecture that will become the basis of the flow for the digital twin platform development. The result of developing the platform is a smart city platform based on a digital twin that can be used to monitor the condition of the city. This platform can be input for users or the community in planning their daily activities and can be decision support to the government in developing the city.

Keywords: digital twin; smart city platform; augmented reality; monitoring; decision support



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1. Introduction

Monitoring and decision support within a smart city platform entails the utilization of technological tools to gather and analyse data from diverse sensors and information sources dispersed across the urban environment. The data gathered encompass several aspects, such as traffic patterns, air quality measurements, noise pollution levels, energy usage, and other related factors [1,2]. By utilizing precise and up-to-date data analysis, municipal authorities can make well-informed decisions to effectively address emergencies, maximize infrastructure development, improve public services, and establish a more efficient and sustainable urban environment. Multiple research studies have shown evidence of the advantages associated with adopting smart cities, including the reduction in carbon emissions, a decrease in traffic accidents, and improving the quality of life for urban residents [3,4]. Furthermore, citizen participation and engagement are critical elements in the context of smart city platforms since they can offer real-time information and foster community involvement to tackle local concerns successfully. Incorporating data-driven decision-making

within a smart city environment exhibits significant potential in augmenting urban living standards and tackling intricate urban issues [5].

The diverse needs of users and stakeholders underscore the need for monitoring and decision support inside the smart city platform. First, it is imperative to acknowledge that citizens need an enhanced standard of living and urban services. It can be effectively accomplished by implementing efficient traffic management strategies, promptly addressing emergencies, and optimizing the allocation of resources. Furthermore, enterprises actively pursue an environment favourable for expansion, relying on up-to-date data to facilitate well-informed decision-making and enhance operational efficiency. In addition, city planners derive advantages from comprehensive data analysis to effectively tackle urban difficulties and strategize for forthcoming urban development. Local governments strive to optimize public services, minimize operational expenses, and foster sustainability by implementing data-driven policies [6]. Public services possess several qualities, including fulfilling societal requirements, establishing and providing public services, coordinating and managing those services, and adhering to a legal framework [7]. Service quality refers to providing prompt and competent user support by a city or organizational operator [8]. In conclusion, implementing monitoring and decision support systems enables the active participation of all stakeholders, fostering transparency and facilitating collaboration. This collective effort contributes to developing a more intelligent, resilient, and sustainable urban environment that benefits the entire community.

Cities have become increasingly smart in the last two decades, using information and communication technology (ICT) to monitor conditions and activities that occur within the city. Data can be generated from various types of activities in the city, such as traffic, transportation, power generation, utility provisioning, water supply, and waste management. The smart city concept is based on the integration of ICT in the life and management of cities and the development of services oriented towards citizens and governments. Various information sources make input for smart cities to develop many services to support citizen activities. Smart city platforms then use this data to improve mobility, environment, living standards, and city governance. However, using these different types of information becomes difficult due to the heterogeneity of the information itself, because sometimes each has its own protocol [3,9]. Consequently, a platform is needed to implement multiple protocols and integrate and visualize information into a single format in real time. This condition is one of the potential areas for the use of digital twins for smart city platforms.

A smart city is a municipality that endeavours to address societal issues by employing ICT solutions facilitated by a collaborative approach that depends on self-governance with several stakeholders [10]. One widely accepted interpretation of a smart city pertains to innovation, focusing not exclusively on information and communication technology (ICT) but also on improving other aspects of urban life, including individuals, governance, transportation, economy, environment, and quality of life [11]. The correlation between the establishment of smart cities and the field of public administration arises from the utilization of ICT to enhance the effectiveness, sustainability, and overall well-being of urban areas. Public administration refers to the actions undertaken by governmental authorities, such as state administration entities, municipal governments, and public law institutions, to fulfil their public responsibilities [12]. Public administration plays a significant role in the formulation, coordination, and execution of many programs aimed at facilitating the transition toward the smart city paradigm. The concept of the right to good administration is manifested through modernizing communication between public administration and individuals, mostly through digital means [10].

Digital twins can monitor and simulate processes or conditions within a city. The results of the simulation can provide feedback to the digital twin model so that it can provide suggestions for fundamental changes in the smart city. Citizens are involved in urban planning, decision-making, and policymaking with digital twins; their input is invaluable. Citizens can engage with all items in the digital twin and report issues or

suggest ideas if this is the case. After pinpointing the precise location of the problem, this complaint or proposal may be reported to the appropriate government agency [9,13,14].

The term digital twin (DT) has become very popular in recent years. DT is a critical technology that enables digital transformation in the industry to digitize physical assets [15,16]. Digital representation of physical assets can allow opportunities to develop new business models or acquire broad ideas [17]. Today, there is hardware that uses Augmented Reality (AR) technology that enables the adoption of new ways of human–machine interaction and information visualization in various use cases [18–21].

A DT provides for depicting physical assets in the digital world, while AR allows for adding digital information to the actual area. Combining these two principles enables a new mode of human–machine interaction, particularly a new mode of situational awareness. Both technologies can help realize seamless user integration into the cyber–physical space [22]. Therefore, situational awareness may be immediately contextualized with physical items instead of relying only on the digital representation of pertinent data. By fusing digital and physical information, users may obtain a complete picture of what is happening in cyberspace and various parts of the real world [18,23]. AR and DT offer an excellent opportunity to sharpen user awareness by directly connecting real-world objects with cyberspace. By seamlessly bridging the gap between the physical and digital worlds, AR and DT provide a fantastic chance to increase users' acuity levels. With AR, data can be shown in any context, and digital twins may be more accurate representations of real-world objects [21,24].

In developing smart cities, the presentation of digital twin data can increase situational awareness. There are five levels of situational awareness, namely the first level of Tuned Out, a condition where individuals are unaware of the surrounding conditions. Second is the level of Relaxed Awareness, a relaxed state but still paying attention. The third is the Focused Awareness level, where caution is observed for potential dangers. Fourth, the High Alert level is a condition where a threat is confirmed, so it is necessary to take action. Finally, a Comatose level, a state of shock, cannot do anything about it. Meanwhile, based on the Endsley Model, situational awareness consists of three levels, namely: Level 1 is perception, which is understanding the status, attributes, and dynamics of elements that are relevant in the environment; Level 2 is comprehension, which is an understanding of current conditions and includes an understanding of the importance of elements and their relation to the purpose of operation; and Level 3 is the ability to project the future of the elements in the environment [25].

Visualizing DT data using AR can make it easier for people to observe the surrounding conditions while casually carrying out activities so that they can be input to plan daily activities. Data visualization is a technique for communicating data or information in visual objects. The goal is to communicate information clearly and effectively to users. Applying AR to the DT concept is primarily driven by a desire to facilitate data availability and enhance comprehension through visual representation. Because of the close connection between DT's digital and physical components, new methods of monitoring and analysing the physical world may be developed with AR. Standard user interfaces, such as web-based, may be used for simulation, analysis, and other tasks; however, AR provides a more natural and productive way to interact with DT data. Some works argue that AR may improve situational awareness by synchronizing data from a DT with its twin in the real world. AR and DT have several potential uses in surveillance, instruction, and damage assessment [18,26].

The AR-DT technology enables users to interact with various pieces of construction equipment, hence increasing operational accuracy and safety. Monitoring and analysing crucial processes is another potential use for combining AR and DT. Some publications provide their efforts with varying degrees of depth to outline the framework and architecture necessary to put AR and DT into practice. There needs to be a best practices framework or architecture for this AR and DT combination. Most researchers only use existing architecture [18,27,28].

Based on the description of the need for a digital twin platform for a smart city, the main problem raised is how to build a digital twin-based smart city platform that can be used to monitor city conditions and can also be used as input for decision-making. Using digital twin technology in developing smart city platforms will improve operational efficiency, the quality of public services, and the environmental sustainability of cities. Therefore, the purpose of this study is to build a smart city platform, where this platform is based on digital twins so that it can visualize data in real time in an environment that aims to communicate information clearly and effectively to users. This environment is also the basis for simulating the city's current condition and development plan. This visualization can be an input for the community in providing advice to the government for the future improvement or development of the city.

2. Literature Review

2.1. Digital Twin Construction

The researchers have provided many definitions of the digital twin (Figure 1). Digital twins are defined by Grieves and Vickers, the two pioneers of the concept, as a set of virtual information constructs that completely describe the manufactured physical product from the micro-atomic level to the geometric macro level. A digital twin is (1) a digital representation of a physical object or process and (2) the establishment of a connection between digital and physical realms to exchange data and information. It exists for real-time monitoring, diagnostics, and prediction, as well as optimizing its dynamic behaviour and keeping tabs on the health of its physical twin [29,30].

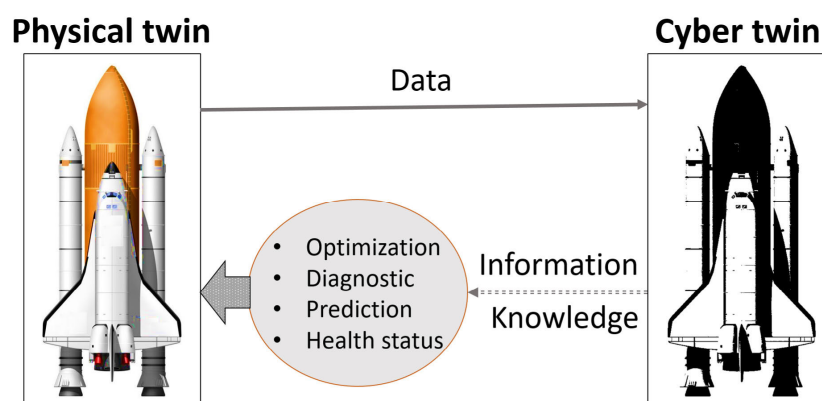


Figure 1. Digital Twin Concept [29].

A physical twin might be a process, a person, a location, an apparatus, or a thing used for a specific purpose. In addition, it is possible to create a digital twin of that, either a limited-function digital twin or a full digital twin that includes the entire behaviour of its physical counterpart [29]. Manufacturing, healthcare, transportation, business, and academia are just a few examples of sectors where digital twins might be used; they could even be used as a novel energy-efficiency solution [20,31].

The primary purpose of the digital twin is to interact with its physical counterparts in the real world. The digital twin can track the current state of a physical object, establish its ideal state, predict its likely future state, and remotely make necessary adjustments to its physical location. In order to specify and mimic the state and behaviour of references to objects that do not yet exist in the actual world, digital twins can be constructed before the usage phase. Afterwards, the digital twin will continue to exist and may be used to archive information about situations that no longer exist in the real world [32,33].

2.2. Maturity Level for Digital Twin

Digital twins have a maturity spectrum, so that with this levelling the effectiveness of the resulting digital twin platform can be mapped (Table 1).

Table 1. Maturity Levels for Digital Twin [13].

Level	Principle	Usage
0	Reality capture (e.g., point cloud, drones, photogrammetry or drawings/sketches)	Brownfield (existing) as-built survey
1	A 2D map/system or 3D model (e.g., object-based, with no metadata or building information models)	Design/asset optimization and coordination
2	Connect model to persistent (static) data, metadata and Building Information Model (BIM) Stage 2 (e.g., documents, drawings, asset management systems)	A 4D/5D simulation, design/asset management, BIM Stage 2
3	Enrich with real-time data (IoT, Sensors)	Operation efficiency
4	Two-way data integration and interaction	Remote and immersive operations; control the physical from the digital
5	Autonomous operations and maintenance	Complete self-governance with total oversight and transparency

The maturity levels above can guide in determining the needs of the digital twin-based platform. In order to attain a mature level of digital twin implementation, it must be supported by a strong infrastructure. Digital twins require infrastructure that enables the successful use of IoT and data analytics, making digital twins run effectively and according to the goals set [17].

2.3. Smart City Based on Digital Twin

Creating digital twins is seen as a new paradigm for building smart cities. With cutting-edge innovations like the Internet of Things (IoT), big data, cloud computing, and artificial intelligence (AI), smart city infrastructure has progressed from static 3D modelling to the hybrid of dynamic digital technology and static 3D models known as the digital twin. This circumstance gave rise to the idea that digital twins facilitate building of smart cities [34].

The digital twin of a smart city is distinguished by four main features: accurate mapping, virtual–real interaction, software definition, and intelligent feedback. Accurate mapping means that digital twin cities realize comprehensive digital modelling of urban roads, bridges, buildings, and other infrastructure. The digital twin can control sensors in the air, ground, underground, and rivers in physical cities to monitor the city’s operating status and finally form accurate information. Interaction between the virtual and the actual world allows for searching various traces, including those left by individuals and the logistics and cars seen in the actual city. A software definition of a smart city indicates that it is based on a virtual representation of an actual city and uses software to mimic urban residents, activities, and infrastructure. On the other hand, intelligent feedback is an early warning of unintended consequences, conflicts, and urban dangers that may be achieved by means such as strategic planning, design, simulation, and others [34].

In one study, the use of digital twin technology in smart cities can be in the form of city operational management, traffic management, public services, flood monitoring, and other city operations. In Figure 2, several smart cities based on digital twins are illustrated. The application of digital twins in smart cities can be in the form of city operational management in the form of a smart city operation brain. Officials from the city will take the primary roles in the Smart City Operation Center (SCOC), and they could also designate a Chief Operating Officer (COO). The Smart City Operation Brain (SCOB) management structure is modelled after the digital twin city. SCOB’s primary roles include (1) planning and assessing the general objectives, frameworks, tasks, operations, and management mechanisms of information development across different sectors; (2) contributing to and reviewing the city’s overall design; (3) creating applicable rules, regulations, and bench-

marks; (4) overseeing the city's information resource integration and sharing; (5) keeping an eye on the city's operations, coordination, and unified command; and (6) advocating for an open, big-data-based ecosystem of socially focused products and services [34].

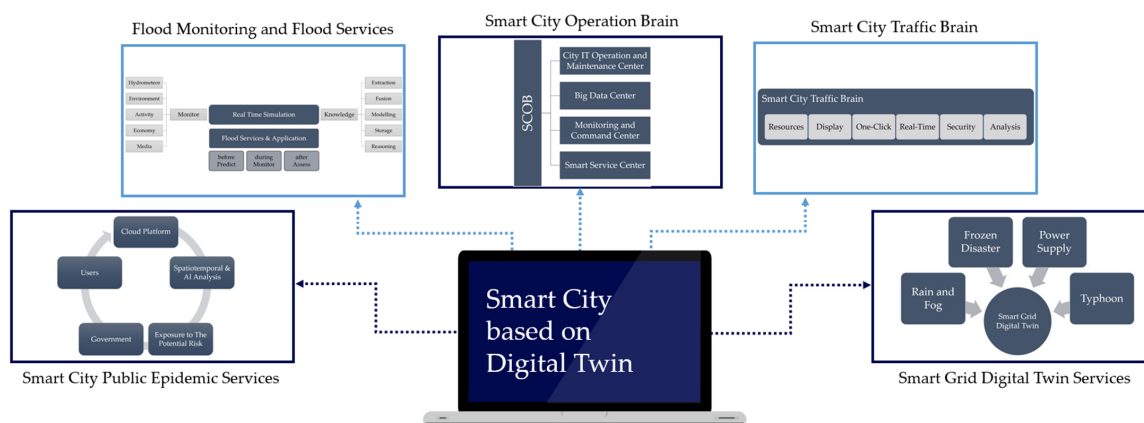


Figure 2. Smart City based on digital twin.

Furthermore, the application of digital twins can also be a smart city traffic brain. Smart traffic systems can be built using holographic technology, time-space analysis, and data mining. The system is designed to integrate dynamic, real-time multinetwork and information traffic resources and connect them with various platform resources such as city alarm systems, police, road condition systems, accident emergency systems, and traffic video systems and display them into the same interface. As for the monitoring function, the digital twin can be applied to flood monitoring and flood services. Three components comprise smart cities' flood monitoring and service apps: big data for flood monitoring, flood maps, and flood service applications. In the context of IoT and with real-time monitoring technology that combines space, air, and earth, flood monitoring big data refers to the monitoring technique of gathering vast quantities of data on flood catastrophes in real-time from an urban and watershed scale. The smart grid digital twin services involve a simulation procedure incorporating comprehensive physical models, real-time measurement data, and historical operational data from power generation systems. Additionally, it integrates diverse disciplines, including electricity, computer science, communication, climate studies, and economics. Smart city public epidemic services can be developed by utilizing various technologies such as cloud data platforms, analytic systems, response systems, and user terminals [34].

2.4. Digital Twin for Simulation

The application of digital twin technology in a city can also be in the form of simulating the condition of a city. The simulations carried out include simulations of the construction of skyscrapers, simulations of green spaces, flood simulations, and simulations of congestion. Based on the simulations carried out, the community can provide input on the simulations carried out, whether they agree or not if the simulation conditions are realized in the real world. Here is an example of a simulation that has been circumvented [9]:

Figure 3 shows that section (a) is the current condition of a city, while section (b) is the condition of the city if there is a new building in the terse-but location. Based on this visualization, the community can provide input or opinions on whether to agree with or disagree with the proposed conditions based on the simulation results. Opinions from the community can be input for local governments to make decisions regarding the direction of urban development policies.

Other simulations can also be completed to simulate floods, congestion, and other problems commonly occurring in a city. Accurate simulations should be backed up with accurate and comprehensive data and information. Without all of these, it is not easy

to carry out simulations that are accurate and complete by the conditions of the city. Inaccurate simulations will also result in mistakes in giving opinions or input, impacting decision-making errors.

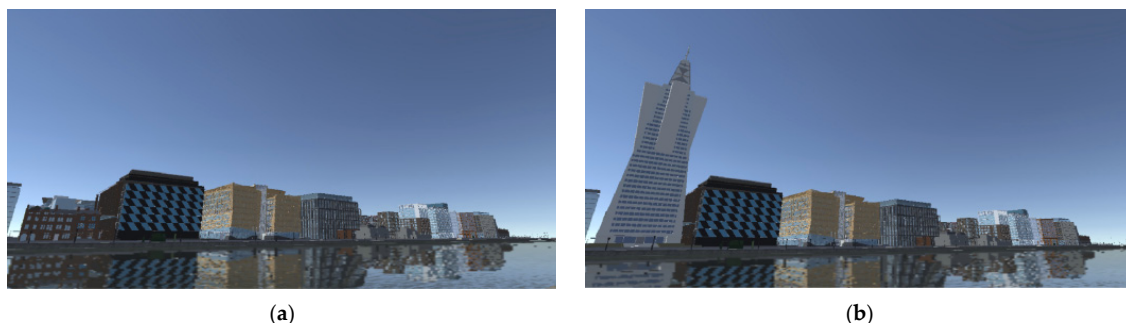


Figure 3. Building construction plan simulation [9]: (a) current digital twin; (b) proposed building digital twin.

3. Methodology

3.1. Platform Development Method

In this study, a digital twin-based smart city platform model was built, which can be a platform for accurately simulating city conditions, involving the community in providing input (feedback), and supporting accurate decisions. This development model is adopted from the digital twin smart city model [9], which divides the digital twin development model into six layers: terrain, buildings, infrastructure, mobility, digital layer/smart city, and virtual layer/digital twin. The development model proposed in this study makes it simpler to build a smart city based on digital twins mapped into four layers, as stated in Figure 4. The built model is as follows:

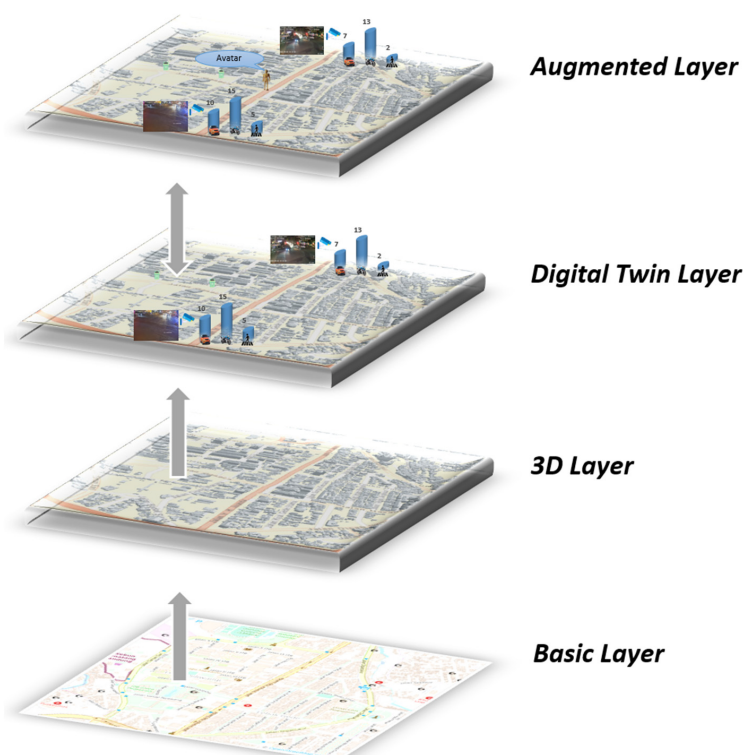


Figure 4. Smart city platform development model.

In the first layer, there is a basic layer that contains basic information about the city, such as street names, building names, location names, and others. The basic layer can be linked to the terrain layer by adding information about the city so that it can use the existing base map. This base screen can be a 2D map retrieved from OpenStreetMap. OpenStreetMap offers current navigation and position data, facilitating the process of route design and exploration. Organizations can leverage OpenStreetMap data to perform market analysis, identify potential client bases, and enhance delivery routes' efficiency. Developers can seamlessly incorporate OpenStreetMap within a digital twin platform, hence augmenting user experiences by providing precise location services. Furthermore, the collaborative nature of OpenStreetMap enables users to actively participate and enhance map details, ensuring the ongoing relevance and accuracy of the information.

On the second layer, there is a 3D Layer. The 3D layer can be equated with the buildings and infrastructure layer because the 3D Layer builds digital objects for buildings and infrastructure. On this layer, a 3D object is built from the area of a city. A 3D object that is built does not necessarily directly build a 3D object but pays great attention to the accuracy, dimensions, area, and coordinates of a physical object in a city. Therefore, this 3D layer uses Shapefile (GIS) data. The data are still LOD Level 1, so some objects are detailed again using the Blender application. This study uses the limited area of Dago, Dipatiukur, and Ganesha in Bandung City, West Java Province, Indonesia. The utilization of Shapefile data offers several advantages, such as enhanced spatial visualization, precise terrain modelling, improved visual analysis, enhanced planning and modelling capabilities, the ability to track changes over time, practical applications in education and communication, informed decision-making, versatile applications, and a more comprehensive geographic contexts. This integration introduces an additional dimension to maps, offering users significant insights and engaging experiences across diverse domains.

In the digital twin layer, real-time data integration is carried out by integrating city 3D, built using sensors, CCTV, IoT, and others installed in the city. The digital twin layer can be likened to the top three layers because this layer already displays real-time data based on sensors installed in the city. At this stage, the process mainly uses the Unity engine. Unity provides numerous benefits compared to alternative programs. The software's cross-platform capabilities facilitate development for multiple platforms and intuitive interfaces. Integrating data into the Unity engine encompasses incorporating external assets, such as images, models, and other relevant elements, into the Unity project. The assets can be arranged, altered, and assigned behaviours through the visual interface provided by Unity.

On the top layer is the augmented layer, where augmenting the digital twin data for several specified points is carried out in this layer. Augmented layer is a new idea proposed with the aim of sharpening the visualization of digital twin data and for development towards the city's metaverse. The augmentation points are major road intersections in the Dago and Dipatikur areas for monitoring the number of vehicles passing through the area per unit of time. There are also several points on Ganesha Road to monitor conditions related to waste locations, micro, small and medium enterprise (MSME) locations, educational places, and places of worship, which at this stage are provided by Avatar to conduct virtual tours. These activities and environments form the basis for developing a more comprehensive future Metaverse.

This study uses the following aspects to measure the virtual experience of using the platform [35]:

1. Interactivity: is related to how fast the data load on the platform, how fast interactions occur, or whether there is a waiting time between performing actions and the platform's response.
2. Challenge: whether operating features within the platform present challenges to users or using them tests the user's capabilities.
3. Skills: regarding operating platforms like other applications in general.
4. Telepresence: is related when operating a platform. It feels like entering the digital world and forgetting the surrounding conditions in the real world.

5. Flow: is related to whether the user feels immersed or has an in-depth experience when operating this platform.
6. Involvement: is related to whether users feel happy and like using this platform.
7. Loyalty: regarding whether users are willing to recommend this platform to others and consider this platform to be the main one to use.
8. Positive Affect: is related to whether the user is happy or unhappy, satisfied or dissatisfied in using this platform.
9. Focused Attention: is related to whether the user's attention in operating this platform can be fully focused and concentrated or cannot be focused and cannot concentrate fully.
10. Vividness: is related to how consistent the information displayed in the virtual environment is with the real-world environment.

Several aspects above can be measured depending on the direction of the platform's focus. The prior approach utilises a measurement aspect to quantify the virtual experience within a 3D virtual reality interactive simulator environment. Therefore, this measurement model is also used to measure how interactivity from the digital twin platform is built.

3.2. Proposed Architecture

The architecture built to develop a digital twin-based smart city platform meets the sensing, understanding, and acting stages concept. The architectural design is as follows (Figure 5):

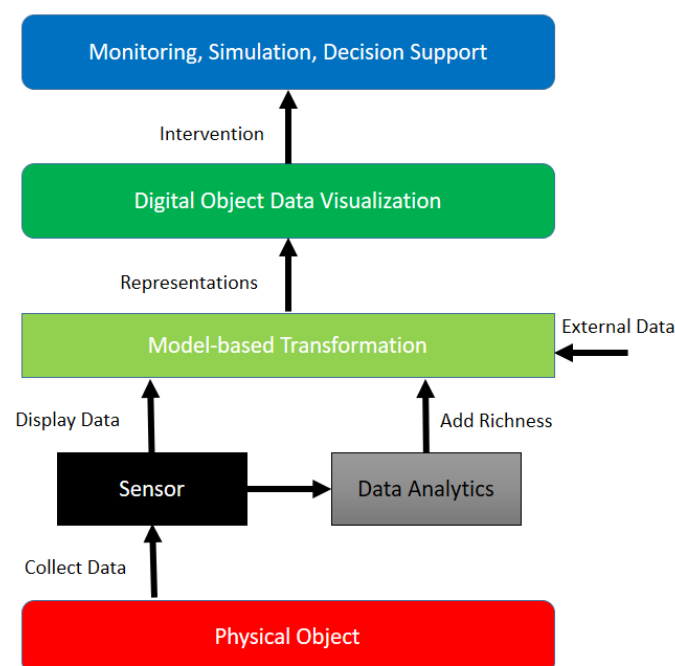


Figure 5. Digital twin architecture for smart city platform development.

The proposed architecture is divided into three stages, namely the sensing, understanding, and acting stages. These three stages are also the basic methods for carrying out this research. The architectural drawing has a physical object at this sensing stage, which is the basis of the built environment. At this stage, data are collected using sensors installed in the physical object environment. The sensors can be IoT, CCTV, or other sensors to detect weather, congestion, and others. Furthermore, from the data obtained, an analysis is carried out to enrich the information on the physical object. It is then displayed together with the sensor results in model-based transformation. External data are needed to sharpen the robustness of the built model.

At the understanding stage (Figure 6), the process of visualizing digital object data is carried out. The purpose of this visualization is to communicate information more clearly and effectively to the user. One way that data visualization can be used is by augmenting information of the resulting digital object. Then, in the acting stage, the resulting platform can be used to carry out the monitoring and simulation process and become a decision support system for users in carrying out their daily activities or for the city government in future city development.



Figure 6. Sensing, Understanding, and Acting Stages.

This developed architecture makes it easier to develop a digital twin-based platform. In the digital twin concept, a platform can already be called a digital twin if it can accurately represent physical objects into digital objects and have real-time connections. In this proposed architecture, to ensure real-time connectedness the presence of sensors are used to obtain data on physical objects. Then, the data are connected in real-time so that the visualization of digital objects can also be real-time. Furthermore, to guarantee the accuracy of the digital model generated based on GIS data, thus guaranteeing the accuracy of the size, dimensions, and coordinates of the physical objects represented in the digital twin platform.

Furthermore, one of the main characteristics of a digital twin platform is that the platform uses software to build a virtual representation of the actual condition of the physical object. Based on the maturity level previously stated, the architecture built allows the use of sensors in the form of IoT, CCTV, and other special sensors so that they can obtain data in real-time so that the maturity level for the platform being built can reach level 3 and make the resulting platform more efficient in its use.

4. Results

4.1. Development of Digital Twin Platform

During the platform development phase, the Unity engine serves as a conducive environment for integrating various data sources such as CCTV data, video analytics, sensors, IoT, and other relevant sources of information. In this research context, it is recommended to utilize OpenStreetMap as the foundational layer for the base map. Moreover, in addition to the foundational layer, three-dimensional items will be incorporated within said region. The 3D model is derived from shapefile data in a Geographic Information System (GIS), enabling the precise modelling of physical items' coordinates, locations, and dimensions. However, it should be noted that the model is currently at LOD Level 1. Subsequently, the Blender application will provide comprehensive descriptions of particular 3D objects, enabling their visualization at LOD Level 2 or 3. The various stages of development are visually represented in Figure 7.

The development stage is carried out following the four-layer platform development model that has been defined previously. Start by creating a basic layer and add a 3D object on top of the basic map. Then, the data that have been obtained are integrated into the Unity engine. Moreover, the top layer carries out data visualization, which can build user interactivity into the platform.

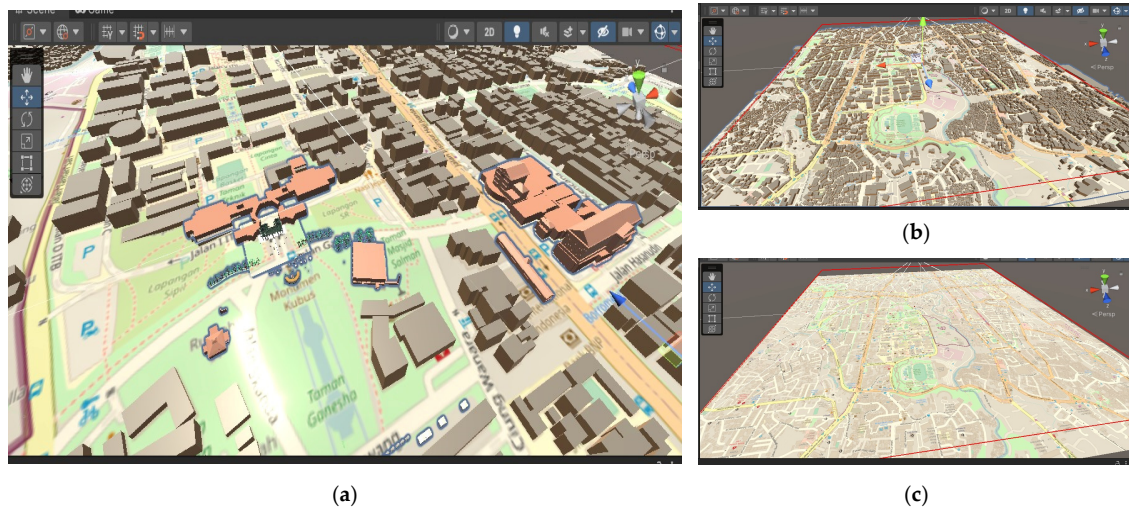


Figure 7. Digital twin city development using Unity: (a) 3D Map; (b) basic Map with 3D Object; (c) basic Map.

4.2. Results of Digital Twin Platform

Digital twins are the basis of the development of this smart city platform. The resulting platform can display 3D of the city and data or information based on sensors installed in the city. One of the displays on this platform is to display the number of vehicles in real-time. It is obtained from CCTV installed at several city intersections. It uses video analytics to calculate the number of vehicles by type, namely cars, motorcycles, buses, trucks, and bicycles. The resulting platform is as follows (Figure 8):

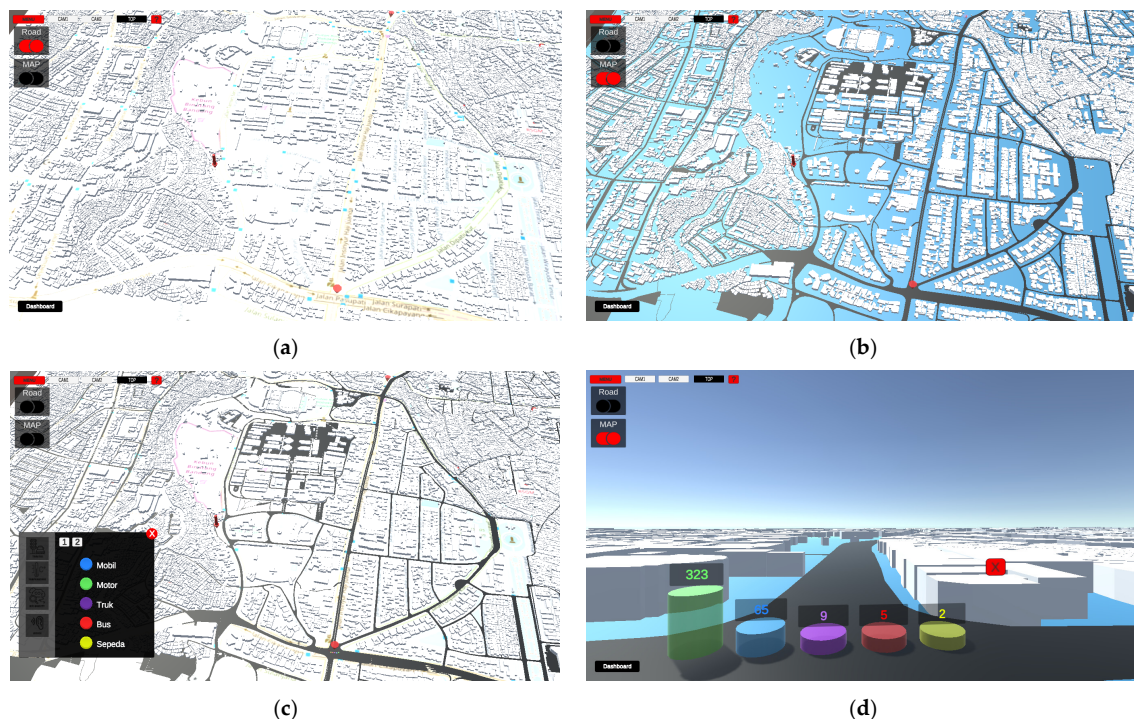


Figure 8. Digital twin cities on restricted areas (Dago, Dipatiukur, and Ganesha): (a) the 3D cities with base maps; (b) the 3D cities without base maps; (c) a 3D city with road and information dashboard; (d) vehicle number information.

On the platform that has been built, three types of views can be seen from the 3D view of the city. First, the 3D city can be viewed with a base map to know each object's location

accurately. Second, the city's 3D can be viewed without a base map, but there is still a road network. Third, the city's 3D can be seen by combining all of them. The platform can display or visualize real-time vehicle count information within a built-in digital twin city or on a digital twin layer. This platform can continue to grow because, in the future, it can continue to add real-time data visualization based on sensor inputs installed in the physical city. With this platform, users or the government can monitor the condition of the city in real-time through this platform without having to be directly in the city's physical location. Therefore, to perfect this platform, it is necessary to complete data or information that can be visualized in real-time. The city government can use the completeness of the data and the accuracy of the data displayed to determine the condition of the city so that it can support accurate decisions in the development of the city in the future.

The resulting digital twin-based smart city platform includes a digital twin with level 3 based on the previously defined maturity spectrum. The digital twin can be enriched with real-time data for operation efficiency. It is evidenced by the resulting platform for obtaining data, one of which uses CCTV. Then, it adds video analytics to display the number of vehicles in real-time or according to conditions in the real world. Suppose there is an addition of sensors or IoT in the future. In that case, it will not change the digital twin architecture built because the platform is a digital environment that can store and visualize data from various sensors.

The digital twin platform that was built can also undergo architectural developments if, in the future, actuators are provided to control physical objects installed in the city's infrastructure. If there is an actuator, it allows two-way interaction so that the resulting digital twin platform can reach level 4. Based on these conditions, where the architecture built allows changes or developments in several sectors, digital twin platforms can continue to increase their maturity level to improve interaction more intuitively and communicate information more effectively.

In the augmented layer, the platform built can be used to interact with complex digital objects. Initially, from the shapefile data obtained, the digital object is only in LOD Level 1. Then, to improve the in-action more intuitively, some selected objects are detailed in LOD Levels 2 and 3. The results can be seen as follows (Figure 9):

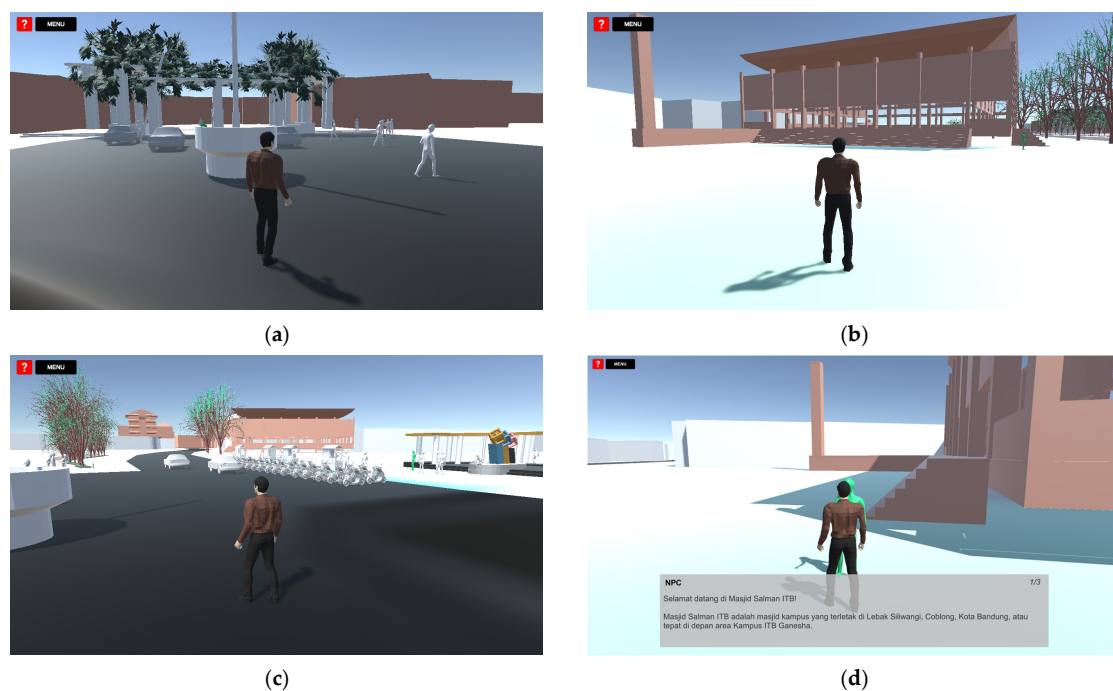


Figure 9. Some 3D objects are detailed: (a) the 3D college gateway objects; (b) the 3D objects of worship place; (c) the 3D objects of parking conditions and road traffic; (d) avatars interact with NPCs.

Multi-object management aims to build interaction between users and digital objects within the platform so that interactions appear to interact with the physical object directly. With this interaction need, an avatar is created as a user representative on the platform. Then, a non-player character (NPC) is provided as opposed to the interaction of the user's avatar. However, the augmented and digital twin layers are not yet in one scene, so the data and information presented in the augmented layer still need to be in real-time. The visualization presented is based on the observation that such conditions are generally in such locations. Making real-time visualizations requires the city infrastructure's readiness to obtain data to display information based on the data obtained. The sensing process to obtain data can use sensors, CCTV, or IoT to detect real-time conditions. This condition is the development of research and platforms in the future.

4.3. Measuring User's Virtual Experiment

The resulting platform is then measured by the user's virtual experience, either using or operating a digital-based smart city platform. The purpose of measuring this virtual experience is to find out how the user's experience is in using the digital twin platform that was built. From the results of user experience, it can be seen to what extent the resulting platform can benefit users and how users provide feedback for future platform development and city development. The aspects measured are interactivity, challenge, skill, telepresence, flow, involvement, loyalty, positive affect, focused attention, and vividness. This study surveyed 57 people to see how users responded to the platform. Of the respondents, 35 people (61.4%) were men, and 22 (38.6%) were women. For the education level, 64.9% are undergraduate graduates, and 17.5% each for post-graduate graduates and still in junior high school/high school. Furthermore, for the category of computer use in a day, it was quite evenly divided, where there were 38.6% stating they used a computer 1–4 h a day, then 29.8% stating they used a computer 5–12 h a day, and 31.6% stating they used a computer more than 12 h a day.

Furthermore, in the virtual experience measurement, the interactivity aspect measured how fast the data load on the platform, so it can be seen how fast the interaction is happening and whether there is a waiting time for the response that appears after taking action. As a result, most respondents stated that the data load on the platform was quite fast. Meanwhile, in operating the platform, the challenge aspect was measured as to whether it provided challenges to users. The result was 49% stating challenging, 32% stating neutral, and 12% stating not challenging. Furthermore, in the skill aspect, most respondents agreed that operating this platform was the same as operating other applications in general. Meanwhile, in telepresence, the user felt like they were entering the digital world and seemed to forget about the surroundings in the real world. The result was that a majority of 74% stated that they strongly agreed (Figure 10).

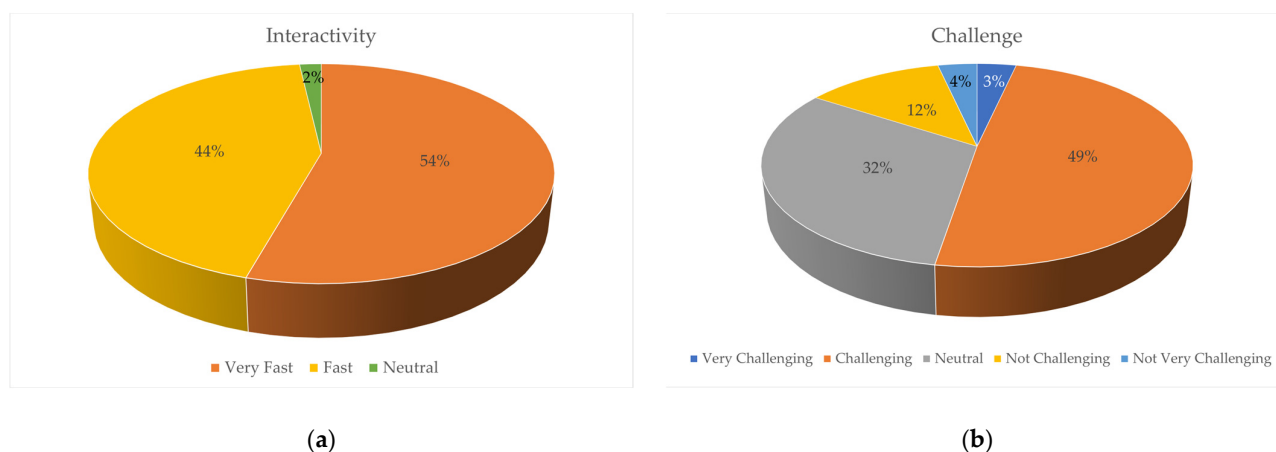
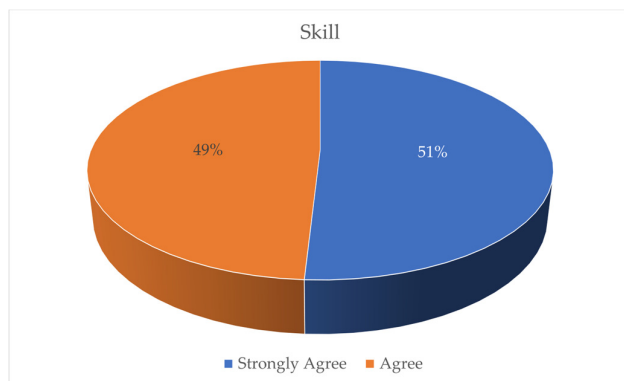
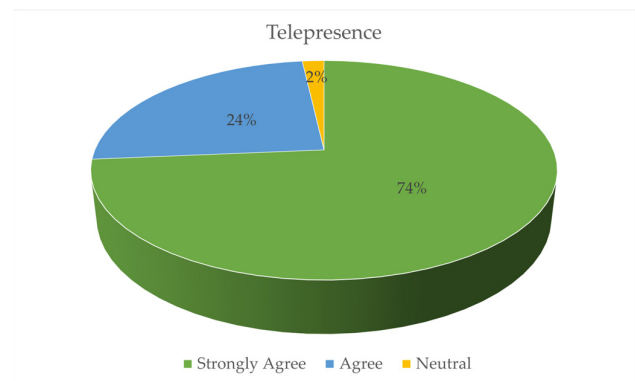


Figure 10. Cont.



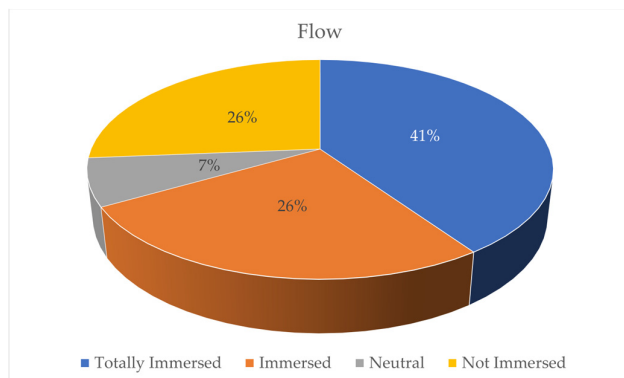
(c)



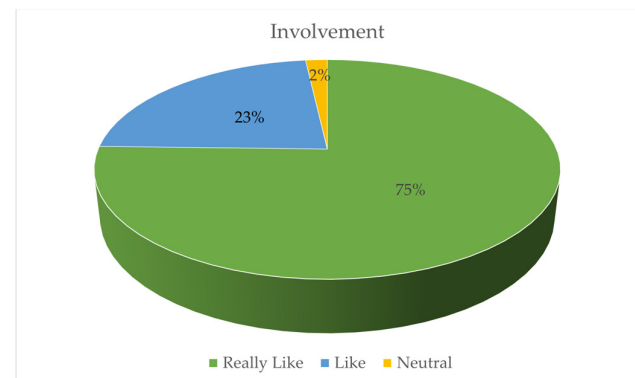
(d)

Figure 10. Result of measurements: (a) interactivity aspect; (b) challenge aspect; (c) skill aspect; (d) telepresence aspect.

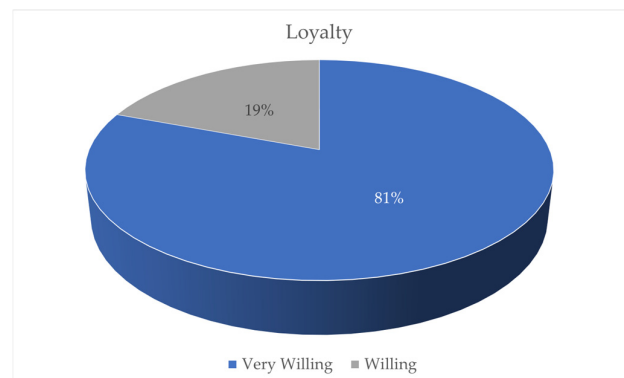
Then, in the flow aspect, 41% felt totally immersed, 26% felt not immersed, and 7% felt neutral. Furthermore, in other aspects, the majority of 75% liked this platform on the involvement aspect, and in the loyalty aspect, 81% were very willing to recommend this platform. In the positive effect aspect, 63% felt very satisfied with the resulting platform (Figure 11).



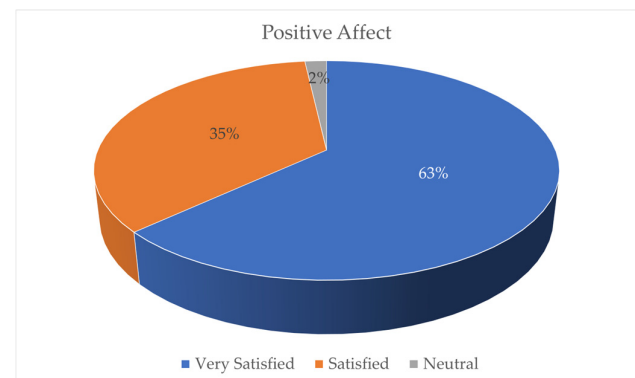
(a)



(b)



(c)



(d)

Figure 11. Result of measurements: (a) flow aspect; (b) involvement aspect; (c) loyalty aspect; (d) positive affect aspect.

In the last two aspects, namely focused attention and vividness, the results were 53% stating very focused and 56% stating the information displayed was consistent between

information in the virtual environment and information in the real environment. The detailed results can be seen as follows (Figure 12):

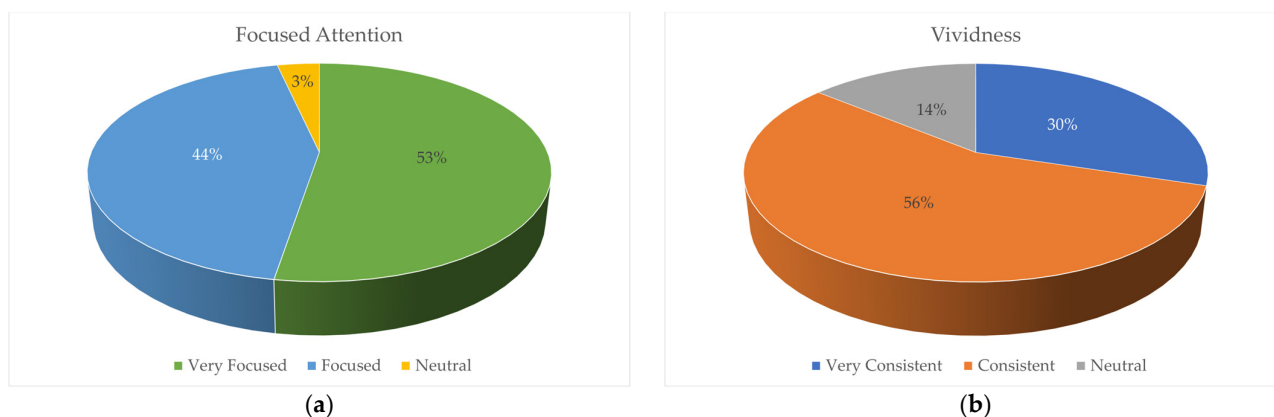


Figure 12. Result of measurements: (a) Focused Attention Aspect; (b) Vividness Aspect.

The findings from the evaluation of the developed platform demonstrate significant efficacy, efficiency, and a range of advantages that are well-matched with the goals of a digitally advanced smart city. The effectiveness of the smart city platform, which utilizes digital twin technology, has been shown through its visualization of the urban area. The phenomenon has increased comprehension of the diverse elements within the urban environment. Integrating augmented data into the platform has greatly enhanced the efficacy of decision-making processes. The utilization of augmented data, obtained from many sensors and real-time analysis, significantly improves the precision in assessing urban trends and future difficulties.

This platform has demonstrated efficient infrastructure management and resource management. The ability to detect difficulties quickly results in a more proactive planning approach and a more efficient allocation of resources. Additionally, the platform's simulation function has facilitated scenario testing, thereby empowering urban planners to proactively anticipate bottlenecks and assess prospective improvements before their actual implementation. The utilization of visualization techniques and the integration of comprehensive data have effectively promoted improved citizen participation and communication. The platform serves as an instrument for municipal authorities to effectively convey urban development strategies, engage citizens in participatory decision-making procedures, and acquire vital input to facilitate iterative enhancements.

Although the platform has exhibited positive results, it is crucial to maintain flexibility to address a smart city's changing requirements and intricacies. With the expansion of the smart cities idea and the emergence of new technologies, it is imperative to consistently refine and integrate innovative features to maintain the platform's enduring relevance in facilitating a digitally transformed urban environment.

5. Discussion

Developing a smart city platform based on digital twins has enabled real-time data visualization. However, some things require special attention. First, this data visualization depends on the readiness of the city infrastructure because with the city's infrastructure running well and entirely, the data can be obtained accurately and run in real-time. Without good city infrastructure, obtaining an accurate digital twin for the city is not easy. The level of city readiness to become a digital twin city can be measured by the completeness of the available city infrastructure, such as sensors, CCTV, IoT, and other sensing tools, and whether the infrastructure has been installed and is still running well.

As previously explained, a digital twin-based smart city has four characteristics, and the built platform already meets these characteristics. First, the digital twin built must be able to map accurately. These platforms use GIS data so that each constructed object is

represented accurately and is scalable for the height, width, and coordinates of its physical object. Second, the built digital twin has been able to display objects according to their location in the physical world so that interactions can be carried out through the platform being built. Third, the digital twin is built in a virtual model according to its physical condition. Fourth, the built digital twin can provide users with feedback or vice versa. In this case, the platform built can be used to simulate city conditions, where the simulation results can contribute to future city development.

At this stage of developing a digital twin-based smart city, this has been carried out by following the development model that has been previously defined. The results of platform development found that there was complexity in integrating and visualizing real-time data into the digital twin platform. It is essential to maintain a digital twin platform that fits the main characteristics that a digital twin platform must have. One of the goals of data visualization on the digital twin platform is to make information representation more effective and a deeper understanding of the data to increase user interactivity on the platform. By cultivating a comprehensive and nuanced comprehension of the user, it becomes possible for the user to provide feedback that can be utilized to advance future platform development. By incorporating extensive user feedback, the decision-making process of city stakeholders can be enhanced with a more focused approach.

6. Conclusions

The development of a smart city platform based on the resulting digital twin can be used to monitor the condition of the city. The platform development stages use a platform development model that divides the development stages into four layers, starting from the basic layer, 3D layer, digital twin layer, and augmented layer. One of the pieces of information visualized within this platform is the number of vehicles by vehicle types, such as cars, motorcycles, buses, trucks, and bicycles. In addition to visualizing that information, this platform can visualize information for temperature, weather, and other conditions. Real-time visualization depends on the readiness of city infrastructures, such as the availability of sensors and IoT, so this platform can be a place to integrate data and information from every sensor installed in the city. Each visualization of this information can be input for users or the community to plan their daily activities and can also support decisions made by the government in planning the city's future development.

This research involved the collection of measurements about the virtual user experience of a platform that was constructed according to pre-established parameters. The platform developed has the potential to enhance users' comprehension of data, hence facilitating a deeper level of user knowledge. This input facilitates users in offering feedback for future platform development. The feedback can afterwards support city stakeholders in making informed decisions about future city development.

Some are of particular concern to future platform development, such as the need for better UI/UX development than what has been produced today, because the primary purpose of visualization is to communicate data or information to users effectively and accurately. With good UI/UX, it is easier for users to understand the latest data or information in the city on the platform. In addition, to make this digital twin platform run well and accurately, it is essential to have the availability of city infrastructure for the sensing process or obtaining real-time data that can be visualized within the platform.

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